

CHANGELOG

Changelog

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

[Unreleased]

Added

- **Foundation distributed-systems library (pkg/)** — a large, growing pure-Go, deterministic, mutation-tested package set. Recent additions include:
 - Consensus/coordination: voting, failconfirm, heartbeatcoalescer, splitbrainalert.
 - Replication/state: mvcc (B-tree time-travel store), antientropy (hinted-handoff + read-repair + Merkle diff), watchmanager (syncd/unsyncd/victim watch groups), crdt/merkle.
 - Scheduling/placement: constraints (Pacemaker 4-type), preempt (value-multiplier), priorityqueue (multifactor aging), workclaim (SKIP-LOCKED), admissioncontrol (N+K reserve), budgetcap, qos, suitability, ewmarank, fallbackchain.
 - GPU/resource & cost: costsched, latencysched, healthmonitor, gpuattest (attestation crypto: challenge/response, proof-of-GPU-work, O(1) spot-check, device-sealing), attestadmit, quantization, local (TCO), pool, burst.
 - Messaging/flow & edge: flowcontrol (K8s APF), idempotent (exactly-once), rebalance (cooperative-sticky), informer (list-watch cache), redundantexec (BOINC trust), edgeregistry, edgeverify, slotmigration, hashslot, healthprobe.
 - Testing/simulation: timefault (clock skew/freeze/monotonic injectors), BUGGIFY in testing/dst, phase7matrix (gap-matrix verifier).
 - This session (34 pure-Go units):
 - bursthysteresis — BurstController with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band (HXC-1504).
 - providerchain — ordered multi-tier provider fallback cascade with retrievable/terminal error classification and injected-clock failover SLA (HXC-1508).

- `gpucatalog` — `SUPPORTED_GPUS` compute-multiplier catalog, `LookupMultiplier`, and attestation-aware Score (attested above un-attested) (HXC-1592).
- `gravaladmit` — HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and `graval.verified` label (HXC-1523).
- `modelrouter` — strategy-to-default-model resolution (latency/throughput/quality/cost) with a typed unknown-strategy error (HXC-1430).
- `gravalverify` — GraVal GPU attestation with a VRAM-ratio ≥ 0.95 gate and a concurrent, race-clean `BatchVerify` pass-rate KPI (HXC-1435, HXC-1436).
- `gepetto` — HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict >0.80 high-water anti-starvation zero-capacity cut-off (HXC-1446).
- `chutesaccount` — Chutes API model-list (TEE/price fields) and /users/me account-balance client over HTTP with `Authorization: Bearer` (HXC-1429).
- `cmd/burst-controller` — utilization-driven binary driving `bursthysteresis`; emits a “burst allocation request” on SPILL and a RECOVER line, runID-stamped (HXC-1531).
- `tieredcache` — hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-demotion + promotion, and per-tier hit stats (HXC-1418).
- `smartrouter` — intelligent `ModelRouter` (latency/throughput/cost/TEE/balanced) that always excludes unhealthy models, with a typed no-eligible-model error (HXC-1539).
- `epochresolve` — config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence) (HXC-1397).
- `balancemonitor` — USD account-balance monitor with a strict low-balance floor warning, over HTTP with `Authorization: Bearer` (HXC-1512).
- `scan` — Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock (HXC-1403).
- `llmfailover` — error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract (HXC-1600).
- `llmadapter` — Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order (HXC-1599).
- `deltacrdt` — standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge (HXC-1409).
- `gputopo` — topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe (HXC-1406).
- `fsresidency` — filesystem residency challenge: per-byte-range `ReadAt` + SHA-256 verification proving a model file is really present on disk; truncated/missing/mismatch all fail (HXC-1572).
- `economics` — `RewardDistributor` multi-token distribution with treasury/reinvest splits and a conservation invariant (`sum(ledger) + treasury + reinvest == total`), plus `GetParticipantROI` (ROI%

+ break-even days, typed unknown-participant error) (HXC-1460, HXC-1461).

- `revenueopt` — `RevenueOptimizer.OptimizeAllocation` greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias; self-contained types (HXC-1456).
- `exportcontrol` — export-control / country-code KYC gate: a controlled GPU (H100/A100) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with `ErrExportDenied`; Tier-1 is allowed (HXC-1482).
- `devicecatalog` — machine-readable 64-device taxonomy with case-insensitive substring Lookup resolving a discovered CPU model (e.g. Rockchip RK3588) to its tier / trust-level / compute-class (HXC-1331).
- `compliance` — EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records, with a SHA-256 provenance hash bound to the actual record content (HXC-1481).
- `hybridkex` — hybrid post-quantum key exchange combining X25519 (crypto/ecdh) and ML-KEM-768 (crypto/mlkem) shared secrets via HKDF-SHA256 (group X25519MLKEM768); a tampered ML-KEM ciphertext yields a non-matching key (HXC-1476).
- `carbonsched` — carbon-aware job placement: chooses the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering (HXC-1480).
- `gpuattest` (additive `multigpu.go`) — multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors (HXC-1573).
- `workloadrouter` — `UnifiedManager.RouteWorkload`: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier that re-ranks confidential offers when the workload requires a TEE (HXC-1455).
- `metrics` (additive `tiermetrics.go`) — `TierMetrics` exposes `gpu_tier_utilization`, `gpu_cost_per_hour`, and `provider_health` Prometheus-text series with deterministic ordering and concurrent-safe setters (HXC-1535).
- `cmd/gpu-pool-manager` — GPU pool manager HTTP front-end: `POST /allocate` returns the tier-correct (lowest-tier healthy, name tiebreak) provider as JSON (503 when none healthy) and `GET /providers` lists providers with health; `httptest-gated`, self-contained (HXC-1530).
- `marketplaceadapter` — `MarketplaceAdapter` interface (`Name/GetCurrentPricing/SubmitWork`) with a `Name()`-keyed dispatch Registry and a `ChutesAdapter` over HTTP (`httptest` fixture, non-2xx rejected) (HXC-1454).
- `cmd/e2ee-proxy` — transparent ML-KEM-768 (crypto/mlkem) + AES-256-GCM encrypting proxy: `EncryptingTransport` seals request bodies and `DecryptingHandler` opens them, so the on-wire payload between proxy and upstream is ciphertext-only while the client receives the correct plaintext; tampered ciphertext is rejected (HXC-1532).
- `e2eebench` — `MeasureHandshakeLatency` benchmarks the `hybridkex` full handshake (median + P95), proving median < 1ms on

- host and enforcing per-iteration key agreement (ErrKeyDisagreement) (HXC-1556).
- `archlint + docs/ARCHITECTURE.md` — the hardened L0–L7 architecture diagram and component map, mechanically enforced: `archlint` parses the component-map table and fails the build if any documented component maps to a package path that does not exist on disk (HXC-1424).
- Wave 67 (5 units, subagent-driven, 5 parallel streams in disjoint packages):
 - `multiraft` — `MultiRaftManager` partitioning cluster state into independent per-shard `go.etcd.io/raft/v3` groups, each electing its own leader and committing independently via per-shard `Propose` routing over an in-process `RaftTransport`; write throughput scales with shard count and a shard re-elects on leader loss while others keep committing (HXC-1387).
 - `stonith` — STONITH fencing: a `FencingAgent` interface with IPMI / AWS-EC2 / Azure-ARM / SBD-shared-disk / NoOp drivers and a `MultiLevelFencer` multi-level fallback, with sink-side fence confirmation; the IPMI real-binary path is an honest SKIP-with-reason when `ipmitool` is absent (HXC-1401).
 - `deviceplugin` — Nomad/K8s-style gRPC device-plugin / GRES fingerprinting framework over real gRPC: plugins register full device fingerprints, the registry parses GRES descriptors, atomically applies fingerprints, allocates by request, and rejects oversubscription (HXC-1407).
 - `internal/gpu` (additive `manager_helixpow.go`) — dual-workload capacity reservation: `ReserveForHelixPoW(fraction) + ChutesCapacity()` reporting only the non-reserved remainder, with a strict `>0.80` Helix-load Gepetto starvation guard that zeroes Chutes capacity (HXC-1447).
 - `chutes` (additive `config.go`) — `ChutesMinerConfig` and `ValidatorConfig` deployment descriptors with `Validate()` invariants (non-empty IDs/hotkeys, `GPUCount>0`, non-negative cost, positive cache size, TEE consistency) (HXC-1439).
- **Security fix (HIGH):** closed a TOCTOU replay-bypass in `pkg/gpuattest.Verify` — the nonce check and consume are now atomic, with a concurrent-replay `-race` regression test.
- Every foundation package ships with tests that fail under mutation of the logic they cover (CLAUDE-1 enforcement) and are gated by whole-tree build/vet `+ -race`.
- Initial project scaffold with Go workspace.
- 29 submodules integrated at project root.
- Directory structure: `cmd/`, `pkg/`, `internal/`, `api/`, `web/`, `scripts/`, `deploy/`, `test/`.

Changed

- **Governance:** CLAUDE-3 / AGENT-2 / QWEN-2 documentation continuous-sync guarantee added (restates and cites Constitution §11.4.106 docs-chain); `.docs_chain/contexts/tracked_docs.yaml` extended to enforce README, CLAUDE/AGENTS/QWEN, CHANGELOG, FOUNDATION_PACKAGES, and CODEGRAPH export-sync.

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Added

- Phase 0 bootstrap: submodule cloning and workspace initialization.